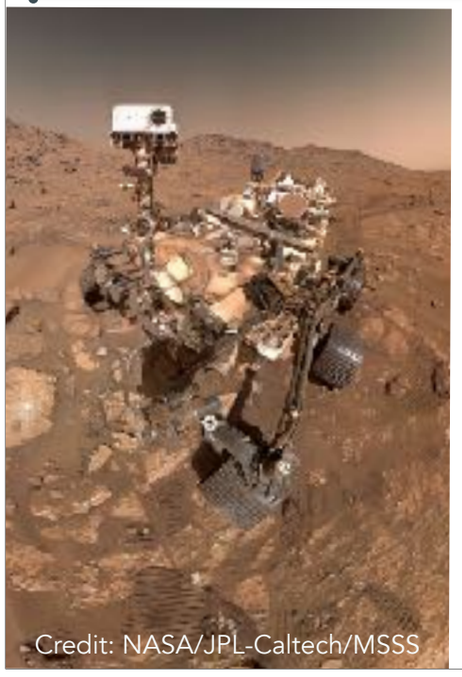


Credit: NASA/JPL-Caltech/MSSS

Intro to Python-Based GIS

Analysis of an Ancient Lake on Mars

Ze-Wen Koh
Shaalin Sehra

Course codes: 12.421/12.621; 16.65J/16.850J

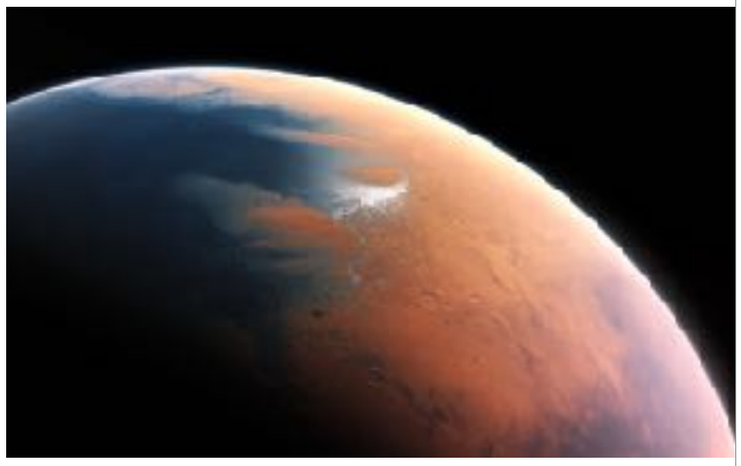
- Yesterday NASA announced that Perseverance has now driven farther than any vehicle on another world, 45.22 km by sol 2006. Today you will map that drive, and the crater it is driving in."

Mars had liquid water

- About 3.5 to 3.7 billion years ago, rivers ran on Mars and filled craters with lakes.
- We know this from orbit: dry river valleys, deltas, and minerals that only form in water.
- Today the air is less than 1 % as thick as Earth's, so liquid water freezes or boils away.
- But we know that the early Mars climate must have been very different, capable of hosting liquid water
  - Say: everything in this story was found by remote sensing, years before any rover arrived.
  -

Mars had liquid water

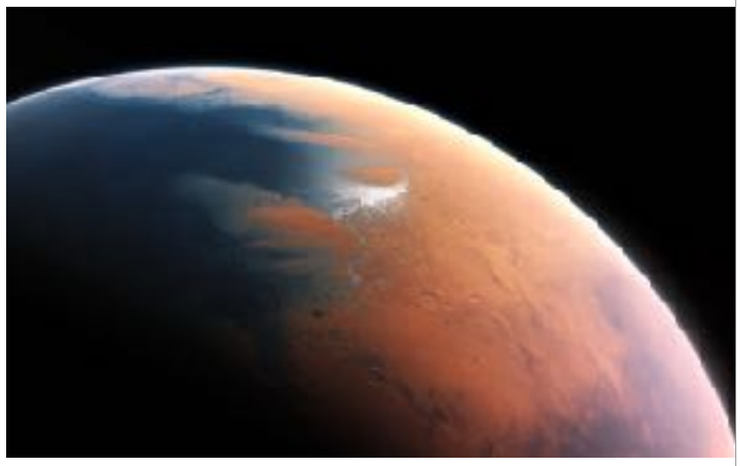
- Leading hypothesis: 3.5 - 3.7 billion years ago, Mars had rivers and lakes



Credit: ESO/M. Kornmesser

Mars had liquid water

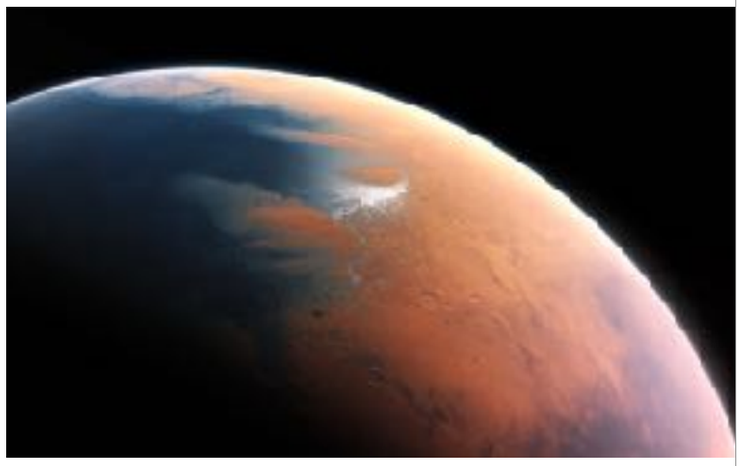
- Leading hypothesis: 3.5 - 3.7 billion years ago, Mars had rivers and lakes
- Remote sensing: Dry river valleys, deltas, minerals that only form in water



Credit: ESO/M. Kornmesser

Mars had liquid water

- Leading hypothesis: 3.5 - 3.7 billion years ago, Mars had rivers and lakes
- Remote sensing: Dry river valleys, deltas, minerals that only form in water
- Today: air <1% of Earth's thickness —> cannot sustain liquid water



Credit: ESO/M. Kornmesser

Crater Lakes: Closed and Open

- Closed basin: water leaves only by evaporation or seeping into the ground.
- Open basin: River going into and out of the lake. the lake fills to the lowest point of the rim and spills over, like an overflowing bathtub. The overflow cuts a canyon, the outlet.
- an Earth example is Lake Bonneville in Utah, which overtopped its rim and drained in a huge flood. Do not say which Jezero valley is the outlet;

Crater Lakes: Closed and Open

- A crater is a natural bowl.
  Rivers flowing in can fill it up,
  forming a crater lake

Crater Lakes: Closed and Open

- A crater is a natural bowl. Rivers flowing in can fill it up, forming a crater lake
- **Closed-basin lake**: River valley goes into lake, water does not flow out



Credit: Kevin Gill

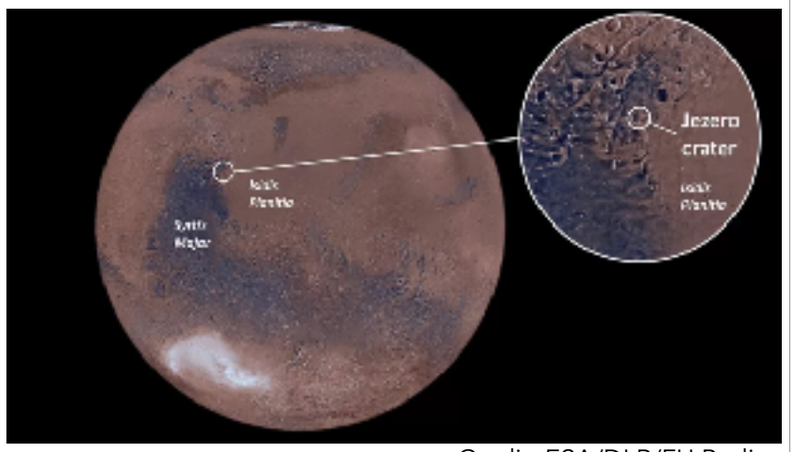
Crater Lakes: Closed and Open

- A crater is a natural bowl. Rivers flowing in can fill it up, forming a crater lake
- **Closed-basin lake**: River valley goes into lake, water does not flow out
- **Open-basin lake**: River going into and out of the lake



Credit: NASA/JPL-Caltech

Jezero Crater and Perseverance Rover



Credit: ESA/DLR/FU Berlin

- • A 45 km crater with a river delta on its western side, clear in orbital images.
- • Perseverance landed here in February 2021 and has collected 29 samples for possible return to Earth.
- Why send rover
- • Orbital spectrometers (CRISM, like your Lab 1 spectrometer but in orbit) found clays and carbonates in the delta, minerals that form in water.
- CRISM mineral colours on CTX, delta in green
- • Deltas trap fine mud fast, which is good at preserving traces of life on Earth.
-

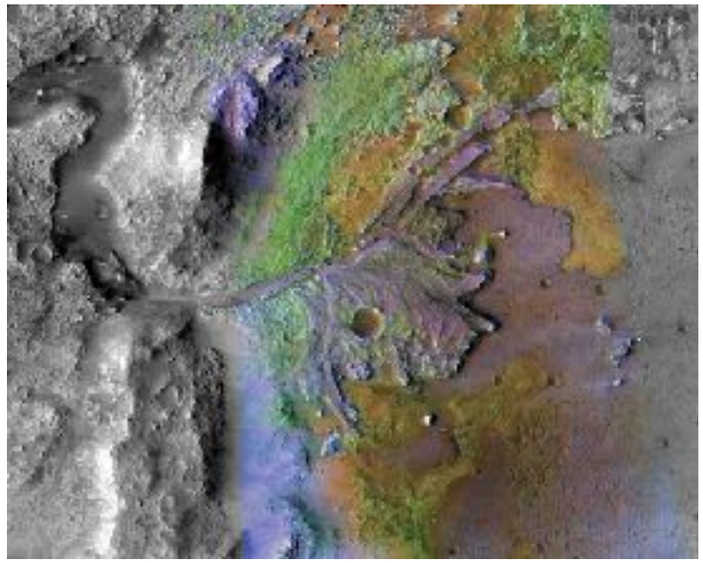
Jezero Crater and Perseverance Rover



Credit: NASA Ames/USGS/JPL/Corrine Rojas

Jezero Crater and Perseverance Rover

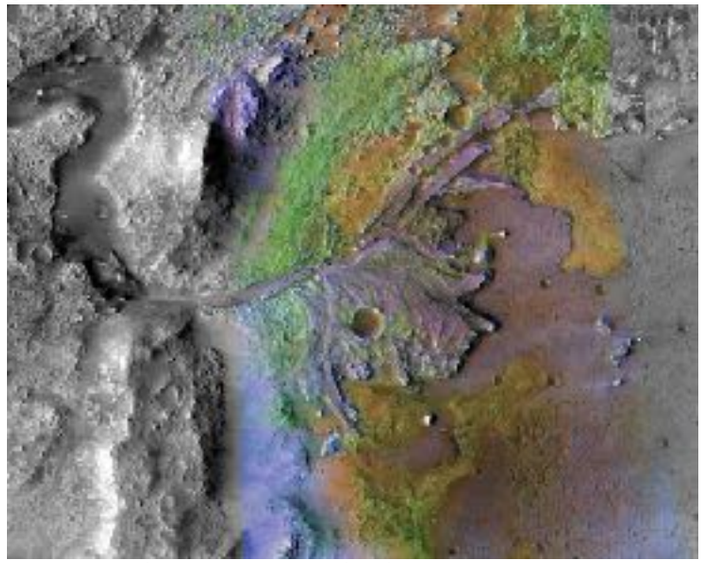
- Crater with river delta —> ancient river and lake



Credit: NASA/JPL-Caltech/MSSS/JHU-APL

Jezero Crater and Perseverance Rover

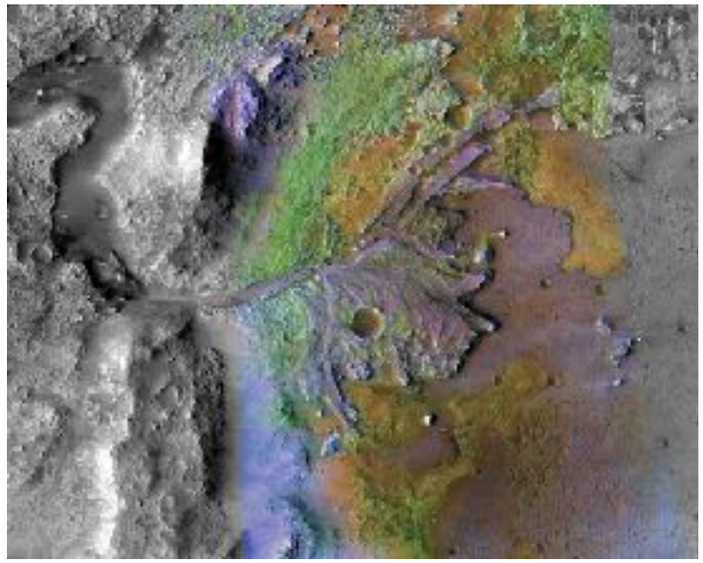
- Crater with river delta —> ancient river and lake
- Why send rover here?



Credit: NASA/JPL-Caltech/MSSS/JHU-APL

Jezero Crater and Perseverance Rover

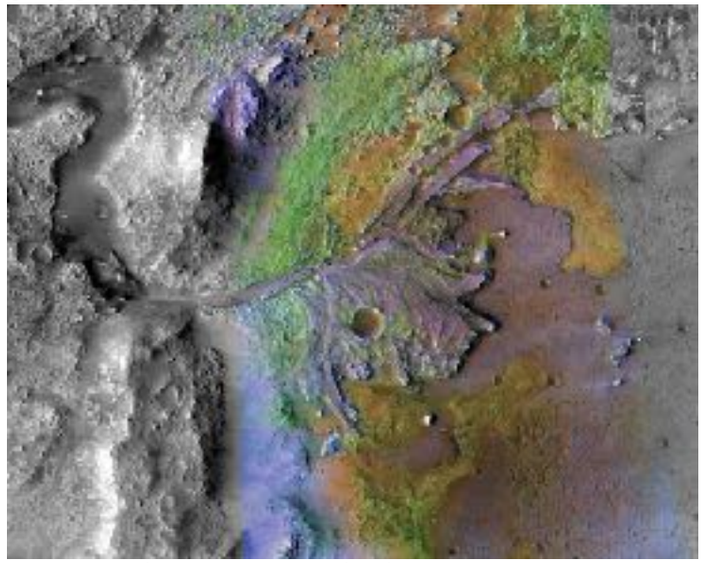
- Crater with river delta —> ancient river and lake
- Why send rover here?
  - **Past water:** Orbital spectrometers found clays and carbonates in delta —> form in water



Credit: NASA/JPL-Caltech/MSSS/JHU-APL

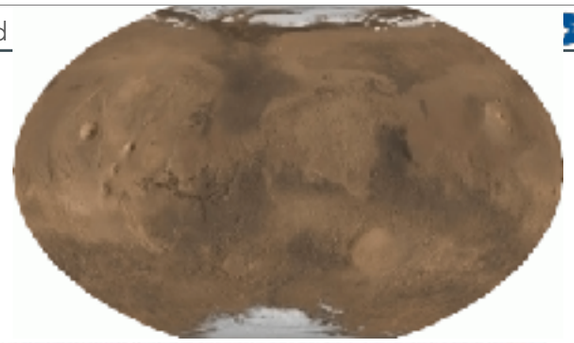
Jezero Crater and Perseverance Rover

- Crater with river delta —> ancient river and lake
- Why send rover here?
  - **Past water:** Orbital spectrometers found clays and carbonates in delta —> form in water
  - **Life**: Deltas trap fine mud fast, which is good at preserving traces of life

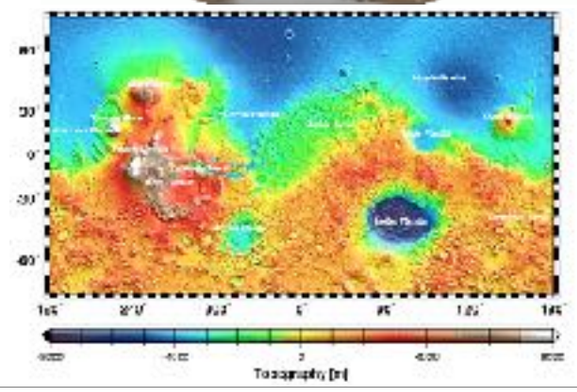


Credit: NASA/JPL-Caltech/MSSS/JHU-APL

CRS, but on Mars



Credit: National Geographic Society, MOLA Science Team, MSS, JPL, NASA

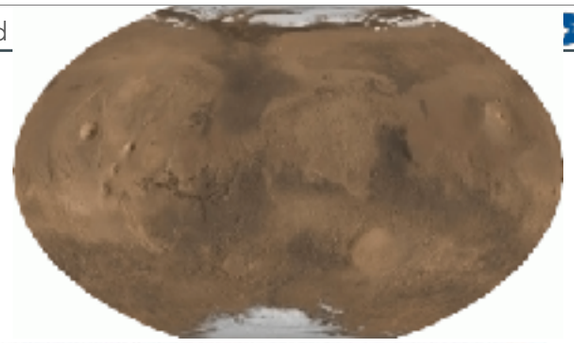


Credit: NASA/ MOLA Science Team

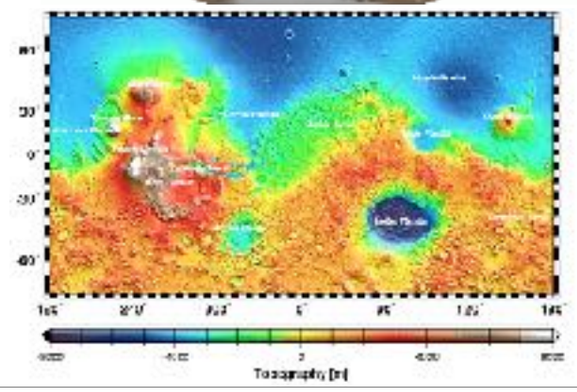
- Same idea as Lab 3: a CRS says what the coordinates mean. But EPSG codes like 4326 and 32619 describe the Earth.
- Mars is modelled as a sphere of radius 3,396.19 km, with longitudes and latitudes on it (IAU codes instead of EPSG).
- There is no sea level.
  - The 0 elevation reference is determined from a gravitational equipotential surface
  - Because Mars doesn't have an even distribution of heavy rocks, mountains, and valleys inside it, its gravity pulls a little stronger in some places and a little weaker in others.
  - • If you walked around the planet following where gravity feels exactly the same strength, you wouldn't be walking in a perfect sphere. You would walk up and down invisible gravitational hills and valleys.
  - • That invisible, slightly lumpy boundary where gravity is perfectly balanced is an equipotential surface. It's the exact same concept we use for Earth's "sea level"—if you filled Mars with water, the water's surface would naturally settle into this exact shape.
- Elevations are measured from this- so most of Jezero is negative, about 2.5 km "below sea level”.
- • A flat map still distorts. In an equirectangular map, the true scale holds only along one latitude, the standard parallel: keep that in mind when you measure areas.
-

CRS, but on Mars

- Different CRS b/c Mars different planet/size



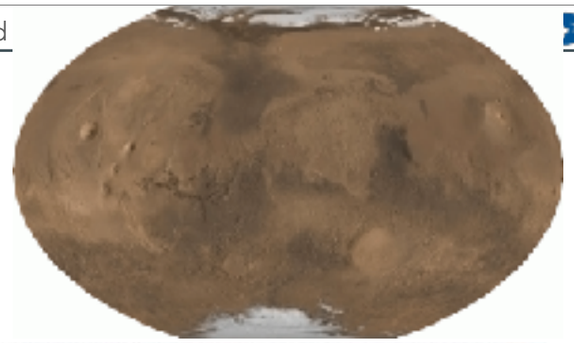
Credit: National Geographic Society, MOLA Science Team, MSS, JPL, NASA



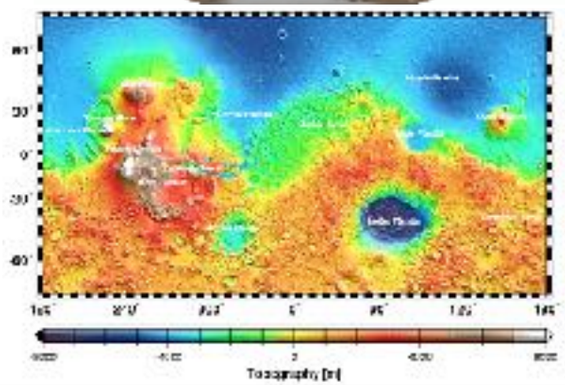
Credit: NASA/ MOLA Science Team

CRS, but on Mars

- Different CRS b/c Mars different planet/size
- **No sea level** —> use gravitational equipotential surface as the reference 0 km elevation



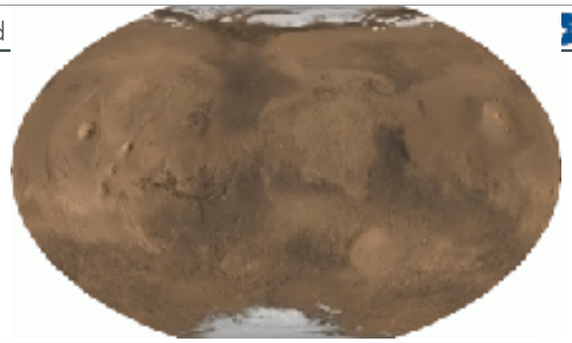
Credit: National Geographic Society, MOLA Science Team, MSS, JPL, NASA



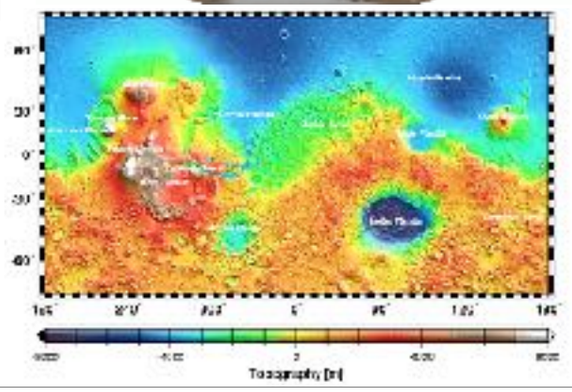
Credit: NASA/ MOLA Science Team

CRS, but on Mars

- Different CRS b/c Mars different planet/size
- **No sea level** —> use gravitational equipotential surface as the reference 0 km elevation
- Jezero crater ~ -2.5 km elevation!



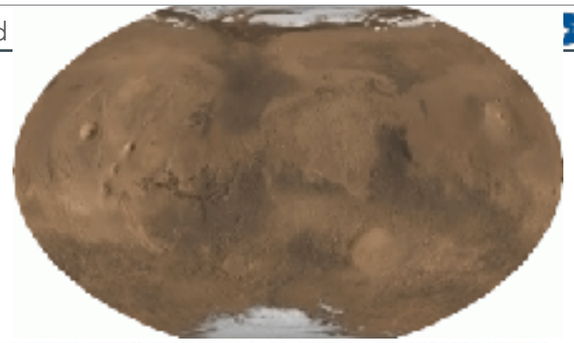
Credit: National Geographic Society, MOLA Science Team, MSS, JPL, NASA



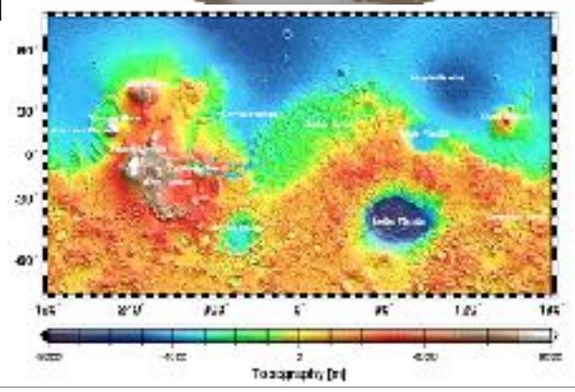
Credit: NASA/ MOLA Science Team

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Credit: National Geographic Society, MOLA Science Team, MSS, JPL, NASA



Credit: NASA/ MOLA Science Team

Todays Datasets

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`jezero_MOLA_HRSC_DEM.tif`	MOLA–HRSC blended DEM, 200 m/px (equirectangular, standard parallel 0°)
`jezero_prebreach.gpkg`	Pre-breach lake polygon (level in `BreachZ_m`), Part 4
`perseverance_waypoints.json`	Perseverance end-of-drive positions (NASA/JPL), Part 5
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- from 200 m pixels to a laser spot under 1 mm, the same resolution question as Labs 2 and 3.

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Line/curve

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wt% of compounds for different locations

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Series of points

Line/curve

wt% of compounds for different locations

Emission Spectra: intensity per wavelength

How CTX DEM was made

- Same parallax geometry as your drone flight in Labs 2 and 3, from about 300 km up instead of 17 m.
- The CTX camera on Mars Reconnaissance Orbiter imaged Jezero twice from different orbits, at different angles.
- Matching features in the two images gives parallax, and parallax gives height: a DEM with 20 m pixels.
- USGS built these CTX maps to guide Perseverance's landing: the spacecraft compared them with its own camera views as it came down.

How CTX DEM was made

- Same parallax geometry as drone flight, from about 300 km

How CTX DEM was made

- Same parallax geometry as drone flight, from about 300 km
- CTX camera on Mars Reconnaissance Orbiter imaged Jezero 2x at different angles



Credit: Gaia Stucky de Quay

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Credit: Gaia Stucky de Quay

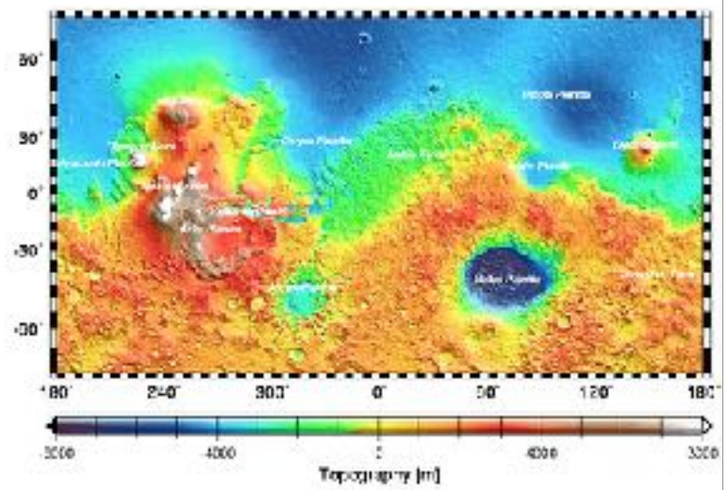
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How MOLA DEM was made

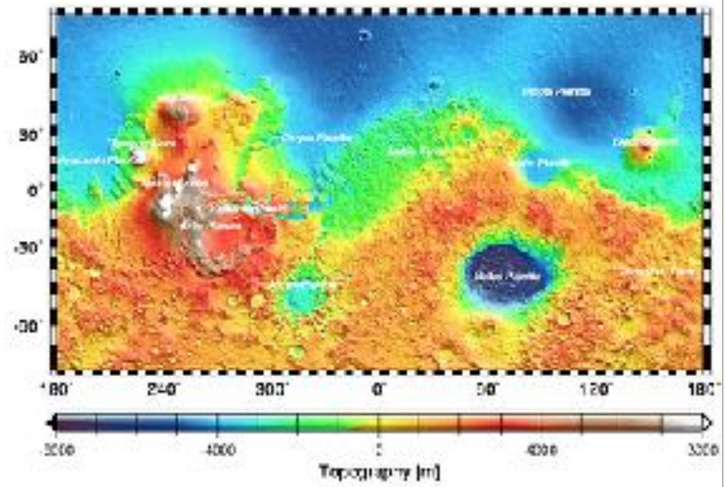


Credit: NASA/ MOLA Science Team

- MOLA, a laser on Mars Global Surveyor, timed laser pulses bounced off the surface: an active instrument, unlike a camera.
- • Blended with stereo images from the European HRSC camera into a global DEM with 200 m pixels.
- • Coarser but covers the whole planet. The CTX DEM is sharper but covers only the region around Jezero.
-

How MOLA DEM was made

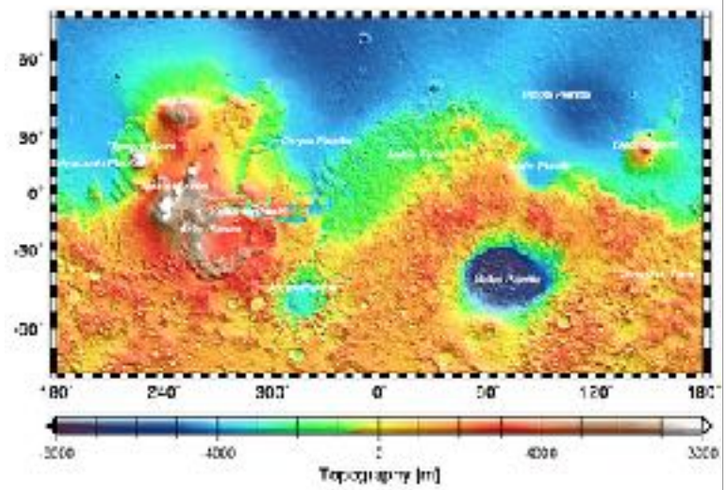
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Credit: NASA/ MOLA Science Team

How MOLA DEM was made

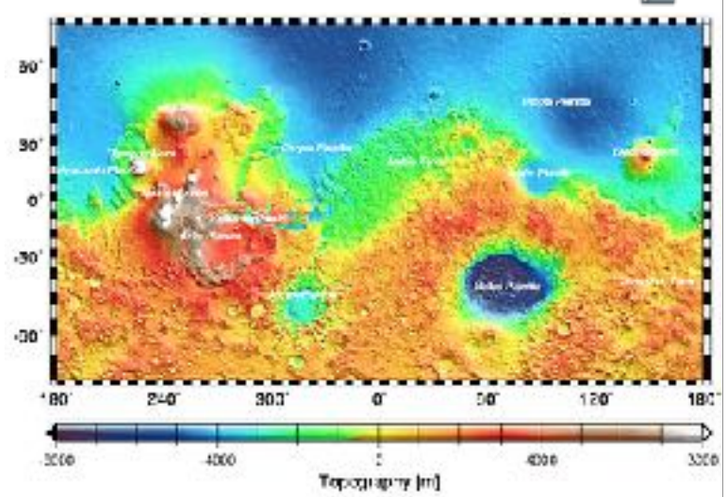
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Credit: NASA/ MOLA Science Team

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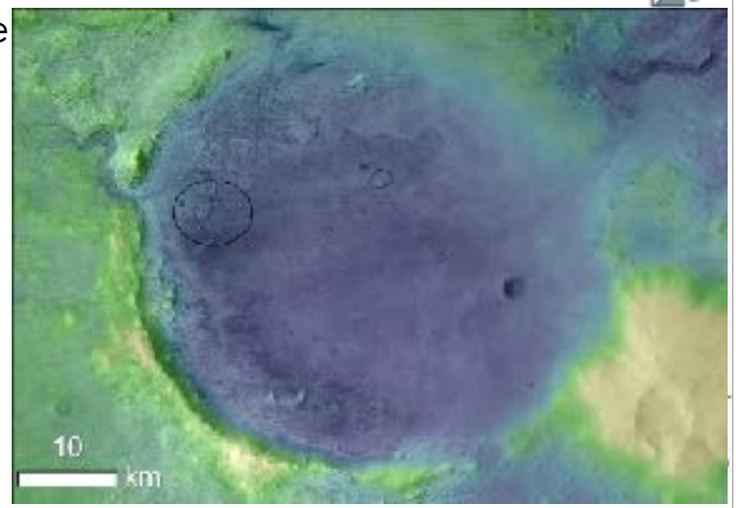
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How MOLA DEM was made

- **MOLA**, a laser on Mars Global Surveyor, timed laser pulses bounced off the surface: an active instrument, unlike a camera
- Blended with stereo images from HRSC, **MOLA/HRSC** is a global DEM with 200 m/pixel resolution
- **MOLA/HRSC:** Coarser but covers the whole planet
- **CTX DEM:** sharper but covers only smaller regions, such as Jezero



Credit: NASA/JPL-Caltech/MSSS/ JHU-APL/ESA

What you will be doing

- Closed basin: water leaves only by evaporation or seeping into the ground.
- Open basin: River going into and out of the lake. the lake fills to the lowest point of the rim and spills over, like an overflowing bathtub. The overflow cuts a canyon, the outlet.
- an Earth example is Lake Bonneville in Utah, which overtopped its rim and drained in a huge flood. Do not say which Jezero valley is the outlet;

What you will be doing

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- Follow Perseverance Rover and its laser unit across the crater